HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented **real-time communication architectures** for enterprise clients, focusing on scalable voice AI and media streaming use cases on the LiveKit platform.
- Collaborated directly with customer engineering teams to integrate LiveKit SDKs and APIs, troubleshooting complex production issues and ensuring seamless deployment of AI agent applications.
- Built **prototype integrations and reference implementations** in Python to accelerate customer development cycles and demonstrate best practices for building and scaling real-time audio/video applications.
- Led technical workshops and architecture sessions with customer developers, translating complex system requirements into actionable implementation plans and providing guidance on infrastructure and deployment strategies.
- Acted as a primary technical point of contact for two enterprise clients during their adoption of the LiveKit platform, facilitating requirements gathering and providing post-sales technical support to ensure successful launches.
- Contributed **Python code and example applications** to the LiveKit open-source ecosystem, enhancing developer experience and showcasing effective platform usage for common agentic AI use cases.
- Containerized LiveKit services and deployment configurations using **Docker** and orchestrated them on **AWS EC2** instances, enabling predictable scaling and simplifying production environment management.

SKILLS

Languages – Python, Go, Node.js, Rust, C++

Cloud & Infrastructure – AWS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes

Real-Time Communication – WebRTC, Real-time Media, Telephony, SIP

AI & Agentic Computing – AI Agents, Voice AI, Conversational Agents, AI Infrastructure

Developer Platforms & APIs – APIs, SDKs, Distributed Systems, Developer Platforms

Methodologies & Practices – Agile, Scrum, CI/CD, Observability, Production Debugging

PROJECTS

Customer Onboarding for Real-Time Agent Platform

- Developed a comprehensive onboarding guide and sample application for a pilot enterprise customer integrating LiveKit for their AI agent communication needs.
- Built **Python integration scripts** to connect their existing data pipelines with LiveKit's event streams, enabling real-time feedback loops for agent performance monitoring.
- Conducted post-onboarding technical review sessions with the customer's engineering lead, addressing integration challenges and documenting best practices for future deployments.

Scalable Voice AI Architecture for Customer Support

- Designed a distributed system architecture using LiveKit and Python microservices to support a high-volume customer support voice AI application, handling concurrent agent interactions.
- Implemented **RESTful APIs** for seamless integration between the AI decisioning engine and the LiveKit media server, ensuring low-latency communication.
- Modeled and tested system performance under simulated peak loads, achieving a **p95 latency of under 200ms** for agent-to-AI message exchange.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)